National Transportation Safety Board
Aviation Accident Final Report

Location:	Hamilton, NJ	**Accident Number:**	NYC08CA042
Date & Time:	11/21/2007, 0001 EST	**Registration:**	N172MG
Aircraft:	Cessna 172N	**Aircraft Damage:**	Substantial
Defining Event:		**Injuries:**	3 None
Flight Conducted Under:	Part 91: General Aviation - Personal		

Analysis

The pilot of the Cessna 172N stated the airplane was in cruise flight when the engine sputtered, and stopped producing power. The pilot then performed a forced landing to trees, resulting in substantial damage, but neither he nor the two passengers were injured. The pilot said he immediately egressed the airplane, checked both fuel tanks, and found both were empty. Examination of the fuel tanks by a Federal Aviation Administration inspector revealed they were intact, and no evidence of fuel, fuel spillage, or fuel odor was identified at the scene. The pilot stated that he departed his home airport with full tanks earlier in the day and recorded 3.2 hours on the hobbs meter. Several stops were made during the day, which involved six takeoffs and climbs to altitude. In a written statement, the pilot added, "Physically checking the fuel tanks for fuel, instead of relying on [the] hobbs meter and an assumed gallons per hour fuel consumption rate, could have prevented this accident."

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be: The pilot's inadequate fuel consumption calculations which resulted in fuel exhaustion.

Findings

Occurrence #1: LOSS OF ENGINE POWER(TOTAL) - NONMECHANICAL
Phase of Operation: CRUISE

Findings
1. (C) FLUID,FUEL - EXHAUSTION
2. (C) FUEL CONSUMPTION CALCULATIONS - INADEQUATE - PILOT IN COMMAND

Occurrence #2: FORCED LANDING
Phase of Operation: DESCENT - EMERGENCY

Occurrence #3: IN FLIGHT COLLISION WITH OBJECT
Phase of Operation: EMERGENCY DESCENT/LANDING

Findings
3. OBJECT - TREE(S)

Factual Information

Pilot Information

Certificate:	Private	**Age:**	37, Male
Airplane Rating(s):	Single-engine Land	**Seat Occupied:**	
Other Aircraft Rating(s):	None	**Restraint Used:**	Seatbelt, Shoulder harness
Instrument Rating(s):	None	**Second Pilot Present:**	
Instructor Rating(s):		**Toxicology Performed:**	
Medical Certification:	Class 2	**Last FAA Medical Exam:**	07/01/2006
Occupational Pilot:		**Last Flight Review or Equivalent:**	
Flight Time:	88 hours (Total, all aircraft), 10 hours (Total, this make and model), 11 hours (Last 90 days, all aircraft), 8 hours (Last 30 days, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Cessna	**Registration:**	N172MG
Model/Series:	172N	**Aircraft Category:**	Airplane
Year of Manufacture:		**Amateur Built:**	No
Airworthiness Certificate:	Normal	**Serial Number:**	17268088
Landing Gear Type:	Tricycle	**Seats:**	
Date/Type of Last Inspection:		**Certified Max Gross Wt.:**	
Time Since Last Inspection:		**Engines:**	1 Reciprocating
Airframe Total Time:		**Engine Manufacturer:**	Avco Lycoming
ELT:	Installed, activated, did not aid in locating accident	**Engine Model/Series:**	O-360
Registered Owner:	ENS Aire, Inc	**Rated Power:**	
Operator:	ENS Aire, Inc	**Operating Certificate(s) Held:**	None
Operator Does Business As:	Philadelphia Flight Academy	**Operator Designator Code:**	

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual Conditions	**Condition of Light:**	Day
Observation Facility, Elevation:		**Distance from Accident Site:**	
Observation Time:		**Direction from Accident Site:**	
Lowest Cloud Condition:		**Visibility**	
Lowest Ceiling:		**Visibility (RVR):**	
Wind Speed/Gusts:	/	**Turbulence Type Forecast/Actual:**	/
Wind Direction:		**Turbulence Severity Forecast/Actual:**	/
Altimeter Setting:		**Temperature/Dew Point:**	
Precipitation and Obscuration:			
Departure Point:	OCEAN CITY, NJ (26N)	**Type of Flight Plan Filed:**	None
Destination:	CROSS KEYS, NJ (17N)	**Type of Clearance:**	None
Departure Time:		**Type of Airspace:**	

Wreckage and Impact Information

Crew Injuries:	1 None	**Aircraft Damage:**	Substantial
Passenger Injuries:	2 None	**Aircraft Fire:**	None
Ground Injuries:	N/A	**Aircraft Explosion:**	
Total Injuries:	3 None	**Latitude, Longitude:**	39.485000, -74.738333

Administrative Information

Investigator In Charge (IIC):	Brian C Rayner	**Report Date:**	12/20/2007
Additional Participating Persons:	Orton Ogborn; FAA/FSDO; Teterboro, NJ		
Publish Date:			
Note:	This accident report documents the factual circumstances of this accident as described to the NTSB.		
Investigation Docket:	NTSB accident and incident dockets serve as permanent archival information for the NTSB's investigations. Dockets released prior to June 1, 2009 are publicly available from the NTSB's Record Management Division at pubinq@ntsb.gov, or at 800-877-6799. Dockets released after this date are available at http://dms.ntsb.gov/pubdms/.		

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The Independent Safety Board Act, as codified at 49 U.S.C. Section 1154(b), precludes the admission into evidence or use of any part of an NTSB report related to an incident or accident in a civil action for damages resulting from a matter mentioned in the report. A factual report that may be admissible under 49 U.S.C. § 1154(b) is available here.